ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 6 kg	m = F/a.
2	**(d)** 200 N	600/3.
3	**(c)** 12 N	Equal to the applied force at the verge of motion.
4	**(b)** 6 N	Change in momentum/time.
5	**(b)** 60 kg	Mass is constant.
6	**(b)** 8 N	Equals applied force.
7	**(b)** 100 N	10/0.1.
8	**(d)** 15 N	Balanced forces.
9	**(b)** 50 N	Balanced.
10	**(a)** 4 N	a = 2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Net force 12 N (1); a = 2.4 m/s squared (1).	1+1
12	Change in momentum 20 kg m/s (1); F = 5 N (1).	1+1
13	Inertia of motion tends to move in a straight line (1); Car turns under him (1).	1+1
14	Moon: 119 N (1); Earth: 700 N (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	a = 5 m/s squared (1); v = 20 m/s (1); s = 40 m (1).	1+1+1
16	Change in momentum 50000 (1); F = 100000 N (1); Weight = 50000 N; force is twice the weight (1).	1+1+1
17	Change in momentum per second = 2 x 5 (1); = 10 N (1); Newton's second law (1).	1+1+1
	OR: Change in momentum 6 kg m/s (1); F = 300 N for 0.02 s (1); F = 30 N for 0.2 s (1).	1+1+1
18	Rolling friction is much less than sliding friction (1); Less force is needed to overcome it (1); So the box moves easily (1).	1+1+1
	OR: Large mass and large momentum (1); Water resistance is small (1); Needs a long time or distance (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	a = 0.5 m/s squared (1); Retardation 1.5 m/s squared (1); Time = 20 s (1); Distance = 900/3 = 300 m (1); Second law used (1).	1+1+1+1+1
	OR: Surface irregularities (1); Depends on nature and normal force (1); f = 60 N (1); Net 20 N (1); a = 1 m/s squared (1).	1+1+1+1+1
20	p bullet = 8 kg m/s (1); Gun gets equal and opposite momentum (1); v = 8/4 = 2 m/s (1); Recoil is opposite to the bullet (1); Third law (1).	1+1+1+1+1
	OR: u = 20 m/s (1); a = 400/800 = 0.5 m/s squared (1); t = 40 s (1); F = 100000 N (1); m = 200000 kg (1).	1+1+1+1+1

Mistakes to avoid:

- Using mass in kg as weight in N (a 5 kg body weighs 50 N when $g = 10$), or saying that the mass of a body changes on the Moon.
- Using the applied force instead of the *net* force in $F = ma$ when friction acts.
- Saying that action and reaction cancel; they are equal and opposite but act on two different bodies.
- Forgetting to convert grams to kilograms (a 20 g bullet is 0.02 kg) in momentum and recoil problems.
- Dropping the negative sign (direction) of recoil velocity, or believing that a heavier gun recoils faster; recoil speed is inversely proportional to the gun's mass.
- Confusing inertia of rest with inertia of motion, or thinking that inertia depends on speed; it depends only on mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 20 N	F = change in momentum / time = 1/0.05.
2	**(c)** 2 m/s squared	200/100.
3	**(a)** 200 kg m/s	mv.
4	**(c)** 6 kg m/s	p = 0.02 x 300.
5	**(d)** 5 kg	p/v.
6	**(d)** 2 : 1	a inversely proportional to m.
7	**(a)** zero	Weightlessness.
8	**(c)** 5 N	30/6.
9	**(b)** 12 m/s	a = 2.
10	**(a)** its weight, downward	Gravity acts always.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gun exerts a force on the bullet (1); The bullet exerts an equal force backward on the gun (1).	1+1
12	Initial 5, final -5 kg m/s (1); Change = 10 kg m/s (1).	1+1
13	m = 0.02 kg (1); p = 2 kg m/s (1).	1+1
14	Weight mg (1); Acceleration g (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	a = F/m = 30/5 (1); = 6 m/s squared (1); v = at = 6 x 4 = 24 m/s (1).	1+1+1
16	Weight = 10 N (1); a = 10 m/s squared (1); Earth is pulled upward with 10 N (third law) (1).	1+1+1
17	Larger mass, larger inertia (1); F = 2 N and 10 N (1); More force needed for greater mass (1).	1+1+1
	OR: Net = sqrt(36 + 64) = 10 N (1); a = 5 m/s squared (1); Direction at tan = 8/6 to the 6 N force (1).	1+1+1
18	Swimmer pushes water backward (1); Water pushes swimmer forward (1); Walking: foot pushes the ground back; friction pushes forward (1).	1+1+1
	OR: Net 140 N (1); a = 2 m/s squared (1); Second law (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Forces: weight, normal reaction, pull, friction (1); Weight and normal reaction balance (1); Net horizontal force 20 N (1); a = 4 m/s squared (1); Direction along the pull (1).	1+1+1+1+1
	OR: Water gains momentum forward (1); Hose receives equal and opposite force (1); Force = rate of change of momentum (1); F = 10 x 6 = 60 N (1); Fireman must hold firmly (1).	1+1+1+1+1
20	a = (20 - 0)/10 = 2 m/s squared (1); F = ma = 1000 x 2 (1); = 2000 N (1); s = (1/2)at squared = 0.5 x 2 x 100 = 100 m (1); Newton's second law (1).	1+1+1+1+1
	OR: Tablecloth: dishes stay at rest (1); Coin drops in glass (1); Vehicle turning: passengers continue straight (1); Inertia of rest and of motion (1); Related to mass (1).	1+1+1+1+1

Mistakes to avoid:

- Using mass in kg as weight in N (a 5 kg body weighs 50 N when $g = 10$), or saying that the mass of a body changes on the Moon.
- Using the applied force instead of the *net* force in $F = ma$ when friction acts.
- Saying that action and reaction cancel; they are equal and opposite but act on two different bodies.
- Forgetting to convert grams to kilograms (a 20 g bullet is 0.02 kg) in momentum and recoil problems.
- Dropping the negative sign (direction) of recoil velocity, or believing that a heavier gun recoils faster; recoil speed is inversely proportional to the gun's mass.

- Confusing inertia of rest with inertia of motion, or thinking that inertia depends on speed; it depends only on mass.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 10 N s	F t.
2	**(c)** 4000 N	24000/6.
3	**(d)** he and the spacecraft are in free fall	Both fall together.
4	**(c)** 6 kg m/s	0.15 x 40.
5	**(d)** 10 m	a = 5; s = 0.5 x 5 x 4.
6	**(d)** 39.2 N	mg.
7	**(a)** Newton's third law	Action-reaction.
8	**(b)** 8 m/s	a = 4.
9	**(c)** 10 kg	W/g.
10	**(b)** large mass	Same force, larger mass.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Depends on the normal force and the nature of surfaces (1); Greater area means less pressure but more contact (1).	1+1
12	Increases the stopping time (1); Reduces the force on the hands (1).	1+1
13	Truck (1); Greater momentum (1).	1+1
14	Net 6 N (1); a = 3 m/s squared (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ball at rest remains until kicked: first law (1); Greater force gives greater acceleration, F = ma (1); The foot feels a reaction: third law (1).	1+1+1
16	Net 10 N (1); a = 5 m/s squared (1); v = 15 m/s (1).	1+1+1
17	Both fall freely with the same acceleration (1); No contact force from the floor (1); Apparent weightlessness (1).	1+1+1
	OR: a = 2.5 m/s squared (1); v = 10 m/s (1); Change in momentum = 20 kg m/s (1).	1+1+1
18	Force 20 N (1); Acceleration 10 m/s squared downward (1); Momentum 40 kg m/s (1).	1+1+1
	OR: Net force is zero (1); No acceleration (1); Remains at rest or constant velocity (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Book's weight on the Earth and the Earth's pull on the book form one pair (1); Book presses the table; the table pushes the book (1); Pairs act on different bodies (1); Balanced forces on the book are weight and normal reaction (1); These two are not an action-reaction pair (1).	1+1+1+1+1
	OR: In vacuum same acceleration g (1); Weight proportional to mass; inertia proportional to mass (1); In air resistance matters (1); Light ball is slowed more (1); Feather and coin in a tube (1).	1+1+1+1+1
20	Initial p = 6 kg m/s (1); Final p = -4 kg m/s (1); Impulse = 10 N s (1); F = 500 N (1); Impulse = change in momentum (1).	1+1+1+1+1
	OR: Same change in momentum (1); Longer stopping time (1); Smaller force (1); Seat belt, helmet, sand (1); Reduces injury (1).	1+1+1+1+1

Mistakes to avoid:

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